1	1	2

⇒ unique pos

1 1 2
1 2 1
2 1 1

3 unique ~ ?

!2

1 1 2
2 1
2 1 1

!1

3 ~

1 1 2

1 1 2
2 1

1 1 2
2 1

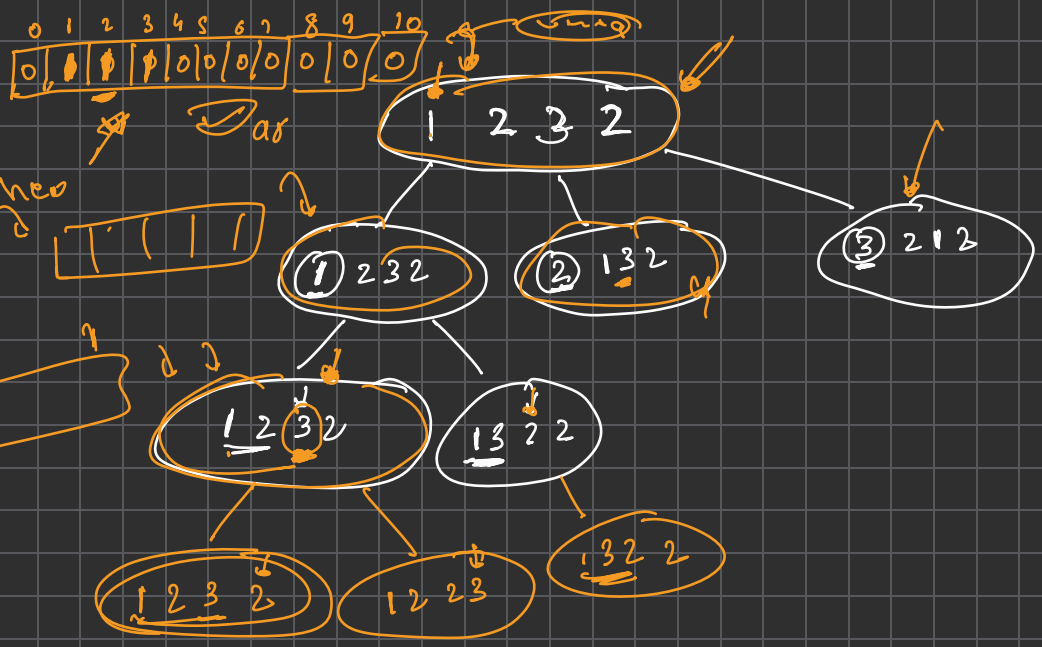
2 1 1
1 1

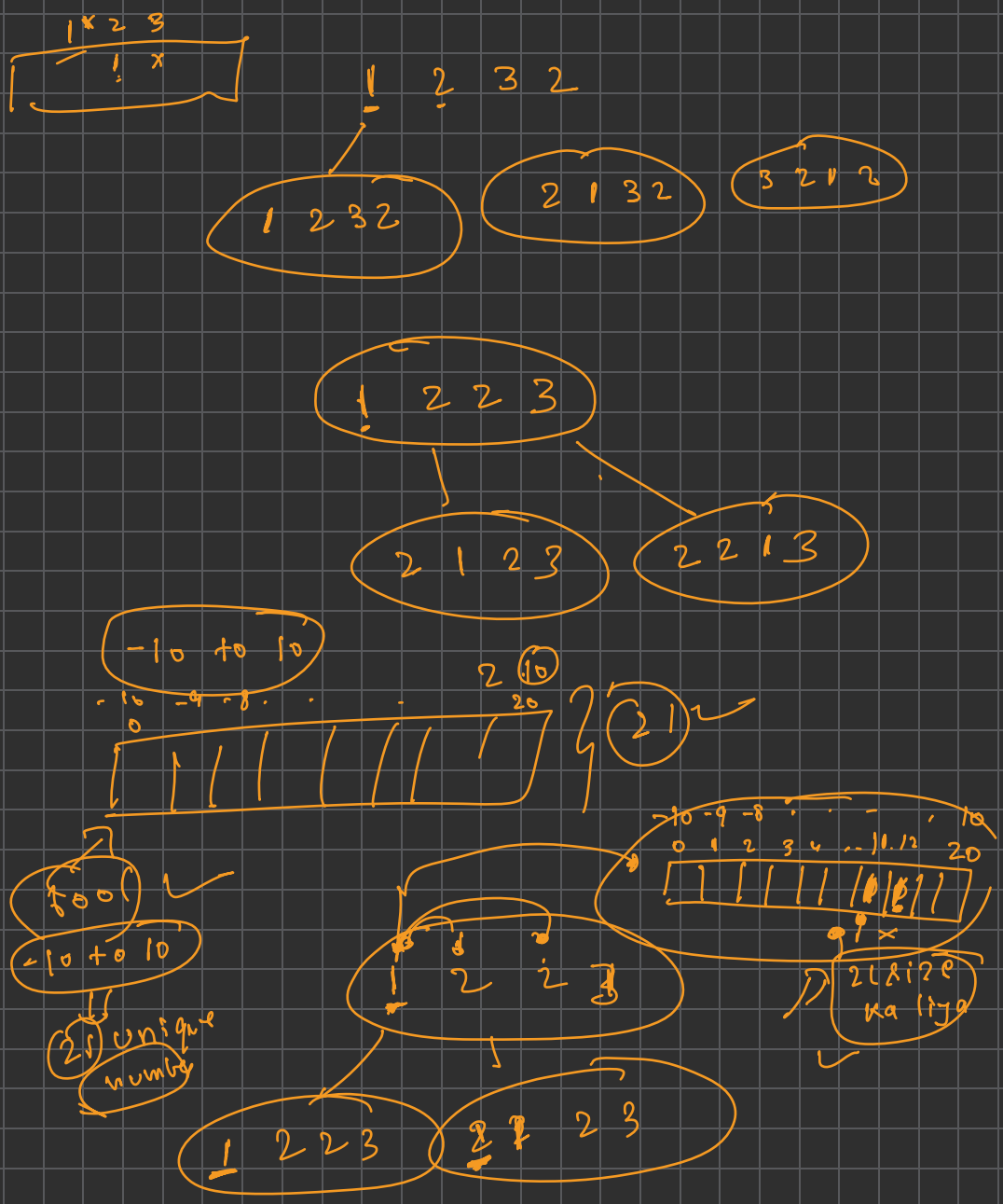
[1, 2, 3]

1 2 3
3 2

2 1 3
3 1

3




1 2 3
1 2 3 2
1 2 3 2
2 1 3 2
3 2 1 2
1 2 2 3
2 1 2 3
2 2 1 3
-10 to 10
2 10
-10 -9 -8 . . . 20
0
21
foo
-10 to 10
21 unique number
1 2 2 3
1 2 2 3
2 1 2 3
-10 -9 -8 . . . 10
0 1 2 3 4 .. 11 12 20
21 size ka liya

[2, 5, 2, 1, 2] , target = 5

[1, 2, 2] . [5]

[1, 2, 2, 2, 5]

[2, 1, 2]

[2, 2, 1]

original array

path =

[1, 2, 2, 2, 5]

(2, 2, 5) 5 -

{2,

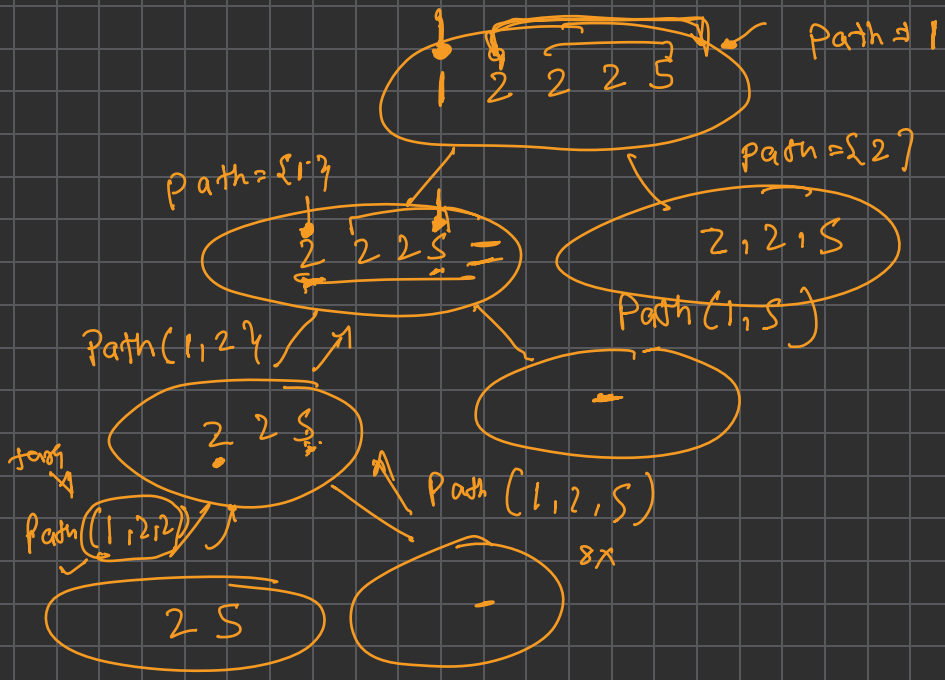
1 2 2 2 5

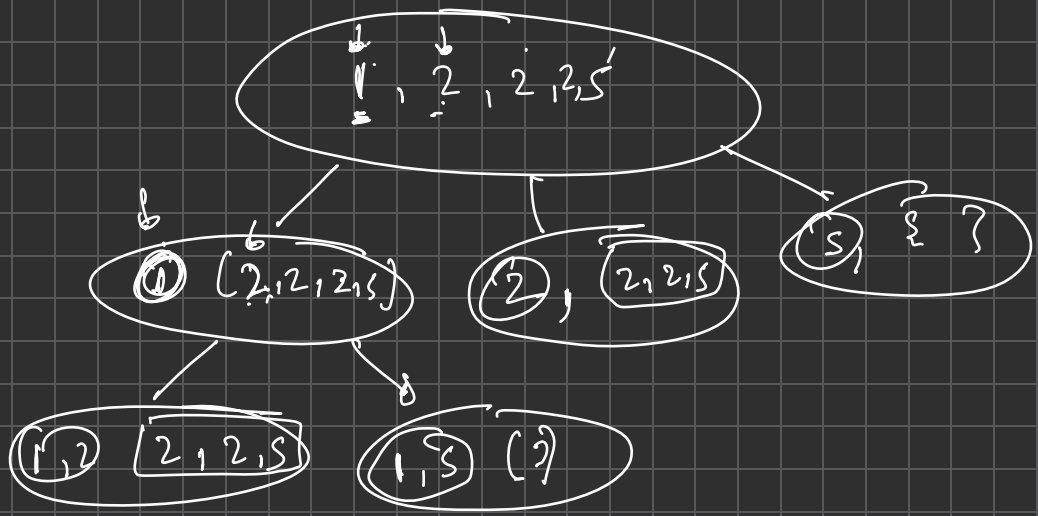
1 2 2 2 5

1 5

1 2 2 2 5

1 2 5


path = 1
1 2 2 2 5
path = {1}
1 2 2 2 5
path = {2}
2, 2, 5
Path(1, 5)
Path(1, 2)
2 2 5
Path(1, 2, 5)
Path(1, 2, 2)
2 5


1, 2, 2, 2, 5
1 (2, 2, 2, 5)
2, 2, 2, 5
5 { }
1, 2 2, 2, 5
1, 5 (2)

0	1	2	3	4
1	2	2	2	5

[1, 2, 2, 2, 5] · target = 5

target = 0

Path

index = 1 to n

target = 4

[1] [2, 2, 2, 5]

target = 3

[2] [2, 2, 5]

[5] []

Path

Path

target = 2

[1, 2] [2, 2, 5]

index = 2 to n

n = 1,

"()" ← string

n = 3 ⇒ (((open

))) ← close

(()))(× total

[()()() (())() ((()))

(()()) ()(())]

n = 3 ⇒ 6 ✓ 3 open, ✓ 3 close

open = 3 , close = 3

(,)

open=3, close=3

(, open=2, close=3

) , open=3, close=2

invalid

invalid

open > close

open < 0

((, open=1, close=3

()(, open=2, close=2

(((, open=0, close=3

(()) , open=1, close=2

((((, open=-1, close=3

(()) , open=0, close=2

×

×

((( )))

(((()) open = 0
close = 1

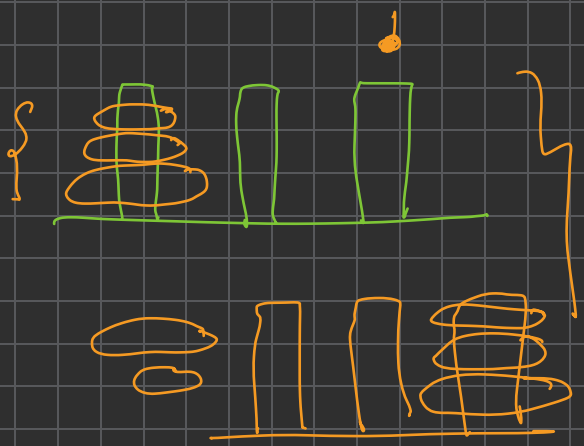
answer will always

(((())) open = 0
close = 0

1

2

TOH



① Small disk ke upar aap badi disk ko nahi rakh

n=3
⇓
7

n=1
⇓

n=2
⇓
3

step

n=3

n=4

step

optim

$n=5$	$n=4$	$n=3$	$n=2$	$n=1$
31	15	7	3	1

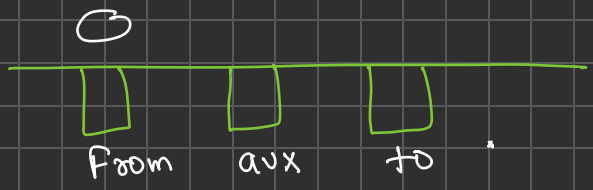
$n=10$

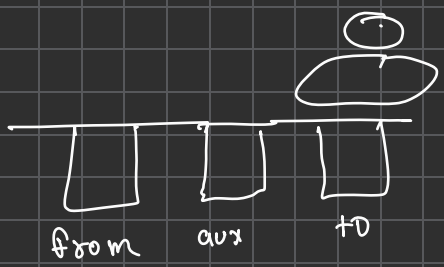
$\Downarrow$

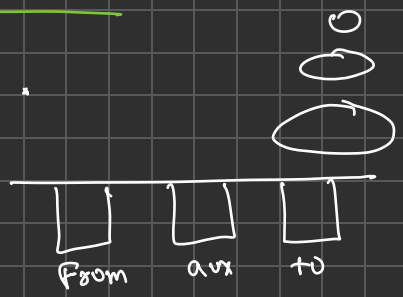
1023

$2^n - 1 \Rightarrow$ ans

step


from
aux
to


from
aux
to

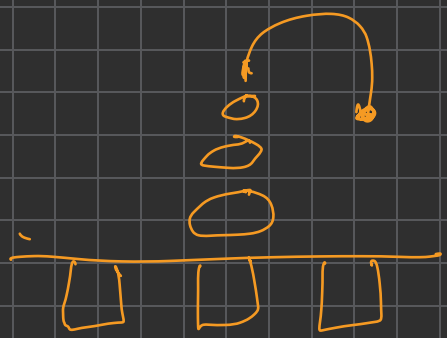

from
aux
to

15 step

$2^n - 1$

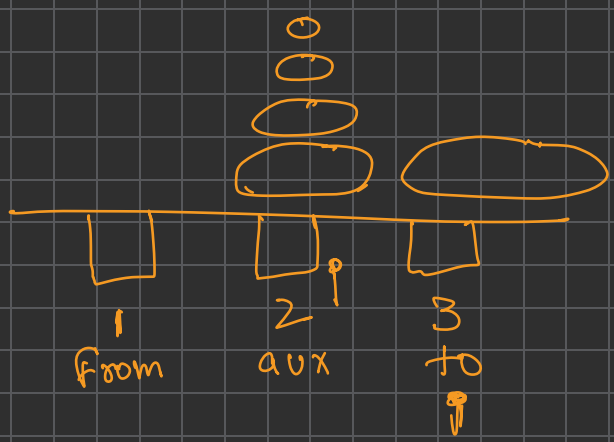
agres

7 step +
1 step
+ 2



15 step
+ 1
15 step

31

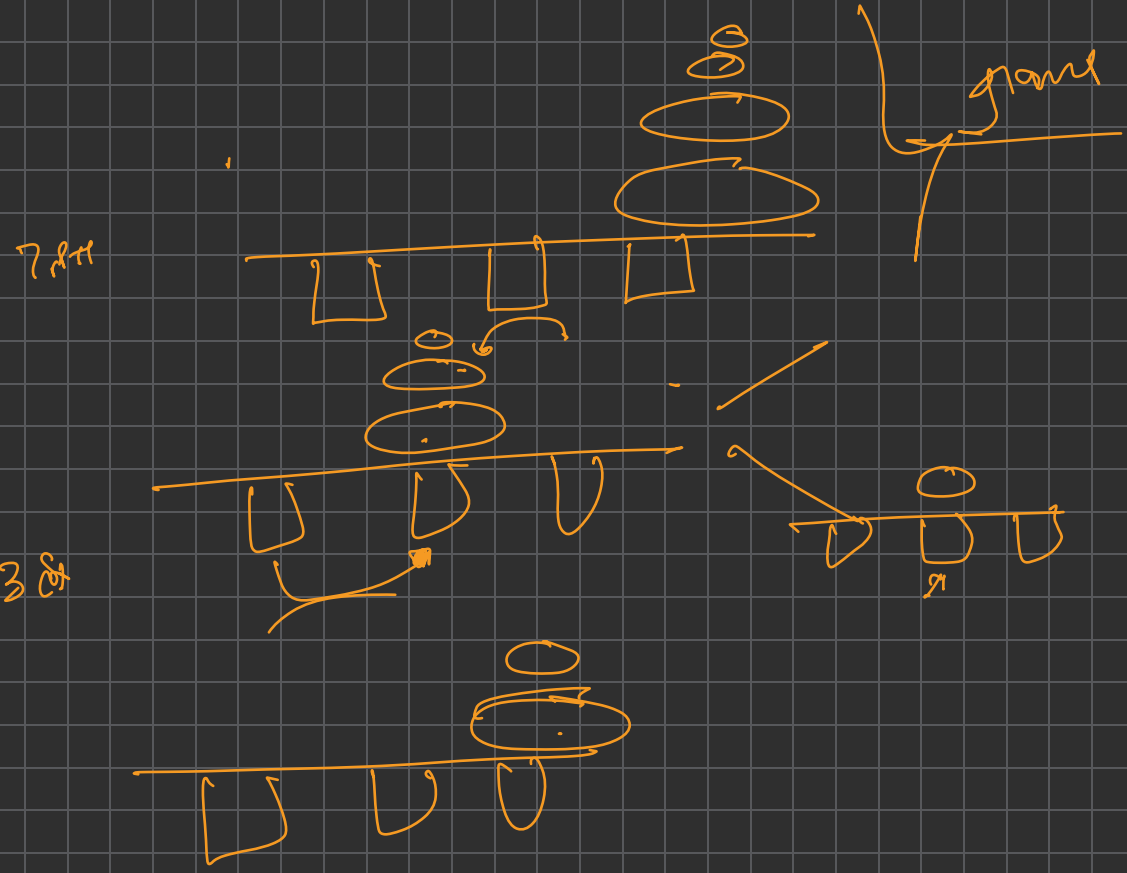


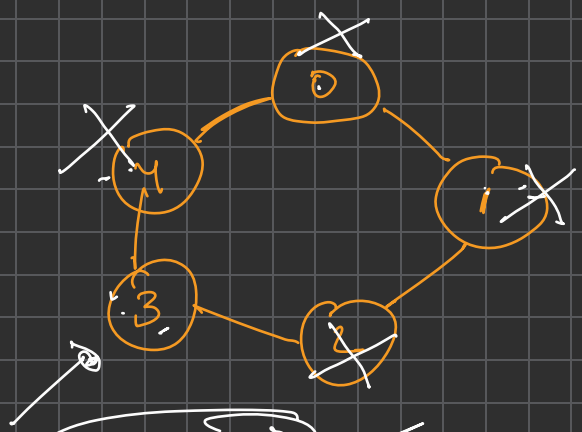
```
TOH( int n, int from, int aux, int to)
{
    if (n == 1)
        return 1;

return  TOH(n-1, from, to , aux)
        + 1
        +
        TOH( n-1, aux, from, to );

}
```

10 disk

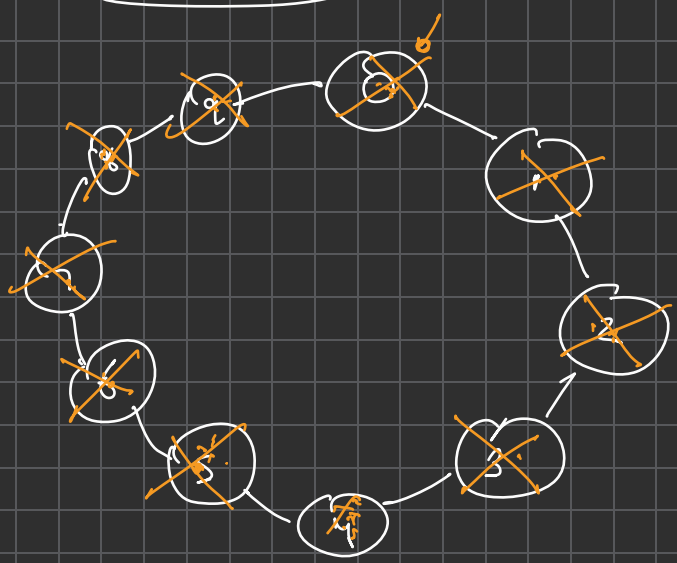

7 8 11
3 8
g



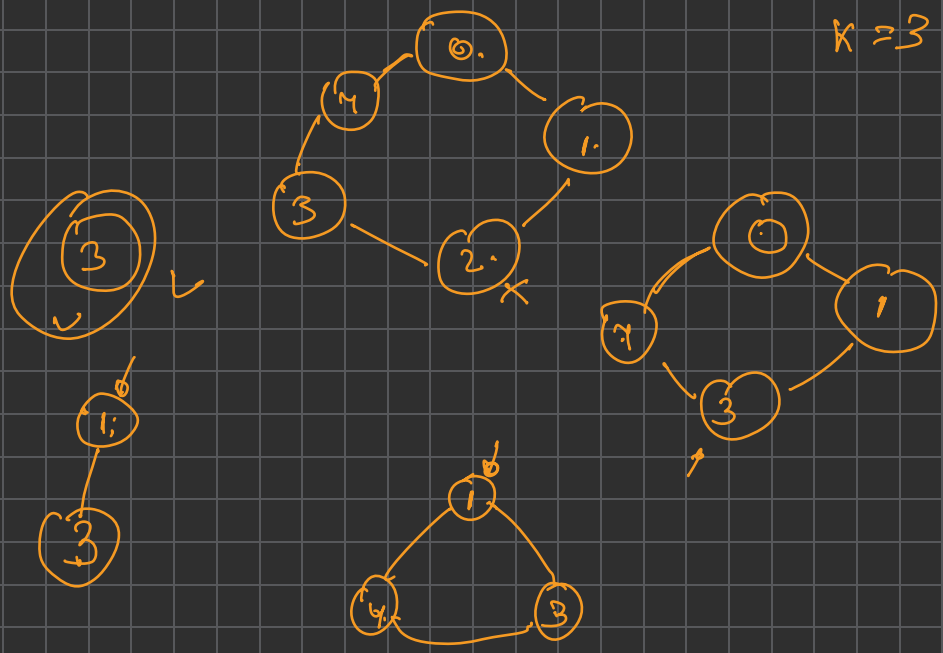
$\boxed{K=3}$

$n=3$

3 winner



$K=4$

4 winner

$K = 3$